



MIRA Miniaturised Space Telescope Lens



BUHO 225 Ground Imaging Platform

in swarm satellites for intelligence gathering, monitoring sensitive borders, and supporting armed forces operations.

In the aerial domain, EON offers optical payloads for drones, eVTOLs, and UAVs, which are capable of detecting movement and identifying vehicles, personnel, or other objects over long distances. Its ground-based systems include pan-tilt day/night and infrared cameras used by the Army, Navy, and Air Force, as well as for crowd and traffic monitoring.

EON Space Labs designs its optical systems end-to-end in-house, leveraging the collective experience of the co-founders in precision optics design, manufacturing, and optomechanical engineering. The design and integration take place



(L to R) Co-Founders Punit Badeka, Manoj Kumar Gaddam, and, Sanjay Kumar

at their Hyderabad facility. At the same time, precision components are manufactured in collaboration with HHV Advanced Technologies in Bengaluru, which specialises in high-precision diamond-turning technology. This strategic partnership allows EON to control the production process, ensuring that the designs stay within India to deliver highly customised solutions while gradually building towards full in-house manufacturing.

The Hyderabad-based startup's aerial and ground products include visible-light and infrared optics embedded in drones and UAVs, with zoom capabilities of 10x to 55x that can observe targets up to 10 kilometres away. Its infrared cameras, with lenses ranging from 19mm to 225mm, can detect humans, drones, or vehicles in low-light and night-time conditions.

Unlike conventional cameras, EON's systems integrate tailored optics with imported sensors, AI, and ML, enabling object detection and tracking in complex environments, such as forests, maritime zones, or defence perimeters.

By designing and producing precision optical systems domestically, the company reduces dependency on foreign suppliers and ensures solutions customised for Indian operational needs. Their systems are validated through metrology

tests, including modulation transfer function assessments, thermo-vacuum testing, and environmental simulations, to guarantee performance under extreme conditions. The company's flagship MIRA-FS telescope is currently undergoing testing, with a flight heritage test in low-Earth orbit scheduled in the coming months.

The startup secured initial funding from HHV Advanced Technologies and MountTech Growth Fund, which supported infrastructure development and access to precision machinery. Its future plans include raising US\$10-15 million to establish in-house manufacturing, reduce lead times, and expand exports.

EON Space Labs' potential customer base includes satellite manufacturers such as Galaxeye, Dhruva Aerospace, Digantara, Kepler Aerospace, and Inspecuity Space Labs, as well as drone and defence companies like IdeaForge, Airbotix, and Arka Aerospace.

Looking ahead, EON aims to strengthen its position across space, aerial, and ground systems by expanding exports, reducing lead times, boosting production capacity, and investing in R&D to deliver customised optical payloads that advance India's technological sovereignty in Earth observation. **EFY**

—Vidushi Saxena